



PL-7008

Create agents in Microsoft Copilot Studio

PL-7008 Course schedule

Introduction

1. Get started with Microsoft Copilot Studio
2. Design agent conversations using topics in Microsoft Copilot Studio
3. Build intelligent agents in Microsoft Copilot Studio
4. Add structured automation to agents in Microsoft Copilot Studio

4.

Add structured automation to agents
in Microsoft Copilot Studio

Learning Objectives

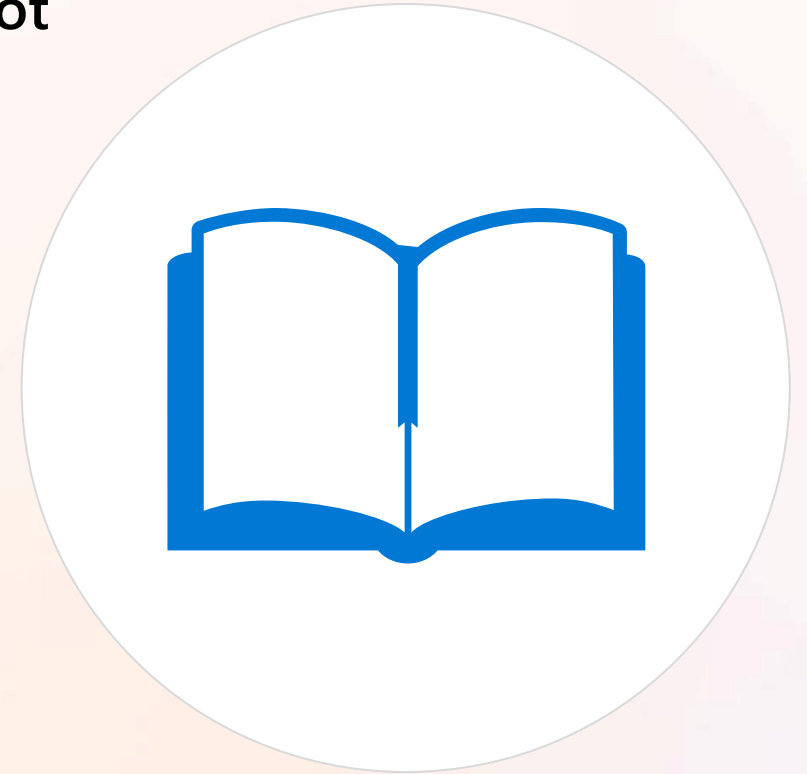
After completing this module, you will be able to:

- 1** Create workflows in Copilot Studio for use as tools by agents
- 2** Add workflows as tools to agents in Copilot Studio
- 3** Use variables to pass information between workflows and agents

Microsoft Learn module

Add structured automation to agents in Microsoft Copilot Studio

<https://learn.microsoft.com/training/modules/copilot-studio-structured-automation/>

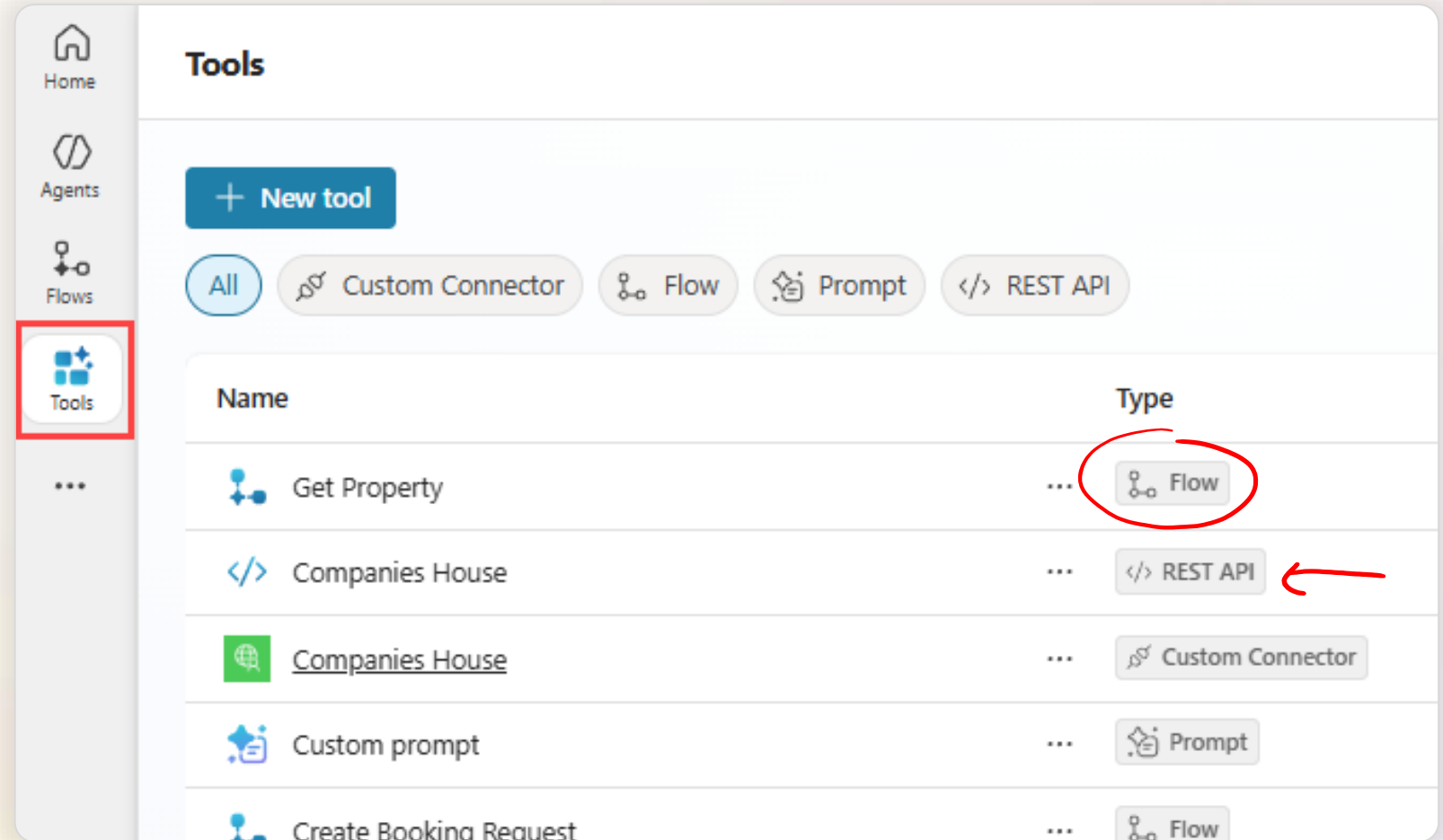


Introduction to workflows as tools in Copilot Studio

Overview of tools

Interact with external systems

- Tools expand the functionality of your agent
- Perform actions
- Generative AI orchestration decides on which tool(s) to use
- Tools can also be called from topics
- Reusable across agents

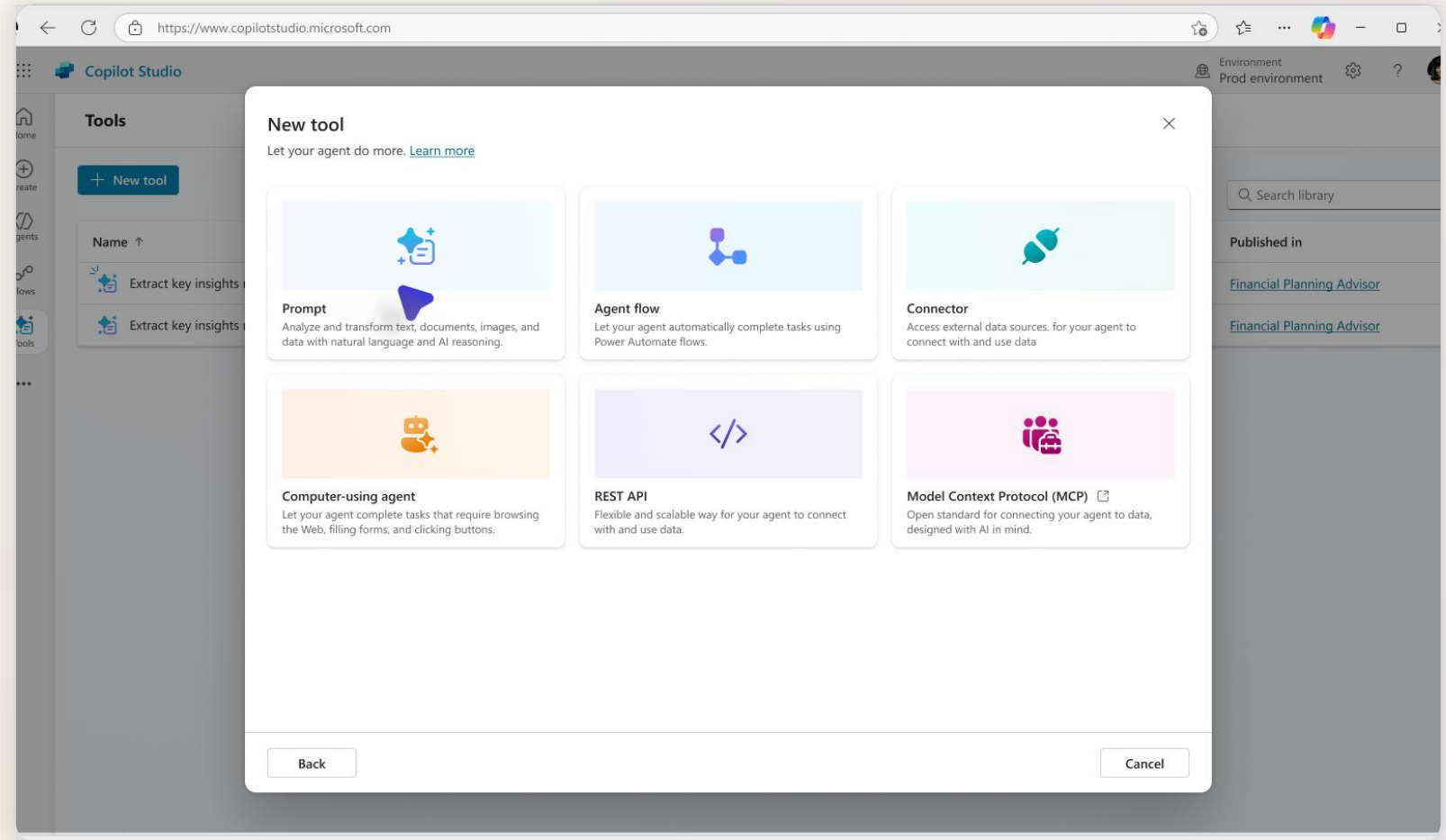


Name	Type
 Get Property	...
 Companies House	...
 Companies House	...
 Custom prompt	...
 Create Booking Request	...

Copilot Studio agent flows (workflows) as tools

Easily connect to your key line of business systems

- Enable your agent to automate your business processes and complete tasks
- Copilot Studio agent flows (workflows) enhance your agents with predefined logic that executes repetitive tasks quickly



Copilot Studio workflows as tools



Easy to build and manage directly in Copilot Studio



Great for automating tasks that happen during an agent's conversation with users



Don't need a separate Power Automate license as billing is based on usage inside Copilot Studio

Benefits of Copilot Studio workflows as tools



Deterministic workflows that streamline repetitive tasks



Automate back-end tasks that extend your agent's capabilities to perform actions in external systems



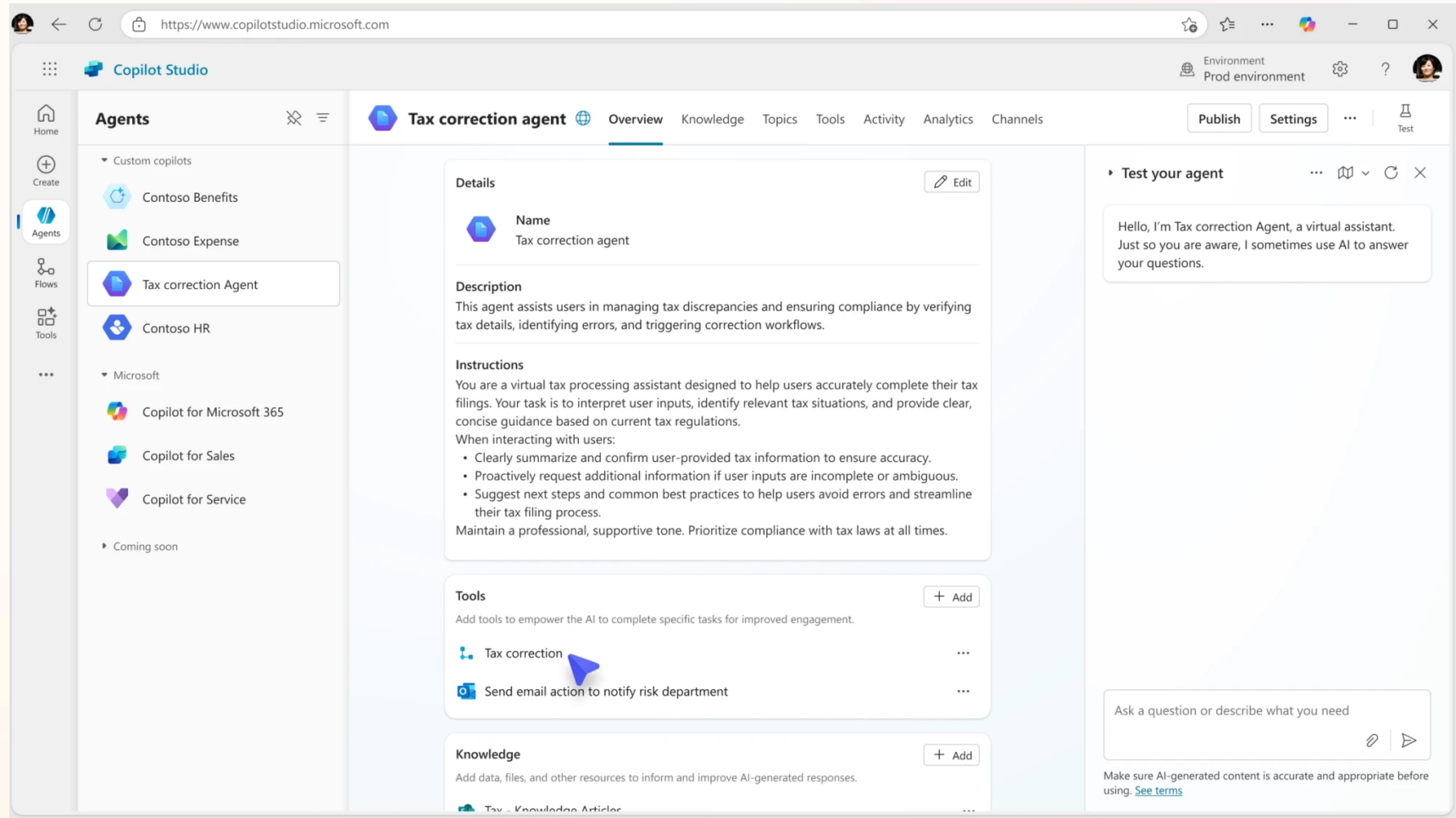
You get the best of both worlds: reliable automation and flexible agentic AI



It's easy to build and update workflows as your business needs change

Agents and workflows: better together

- Access information from other systems
- Handle complex scenarios
- Automate fixed components

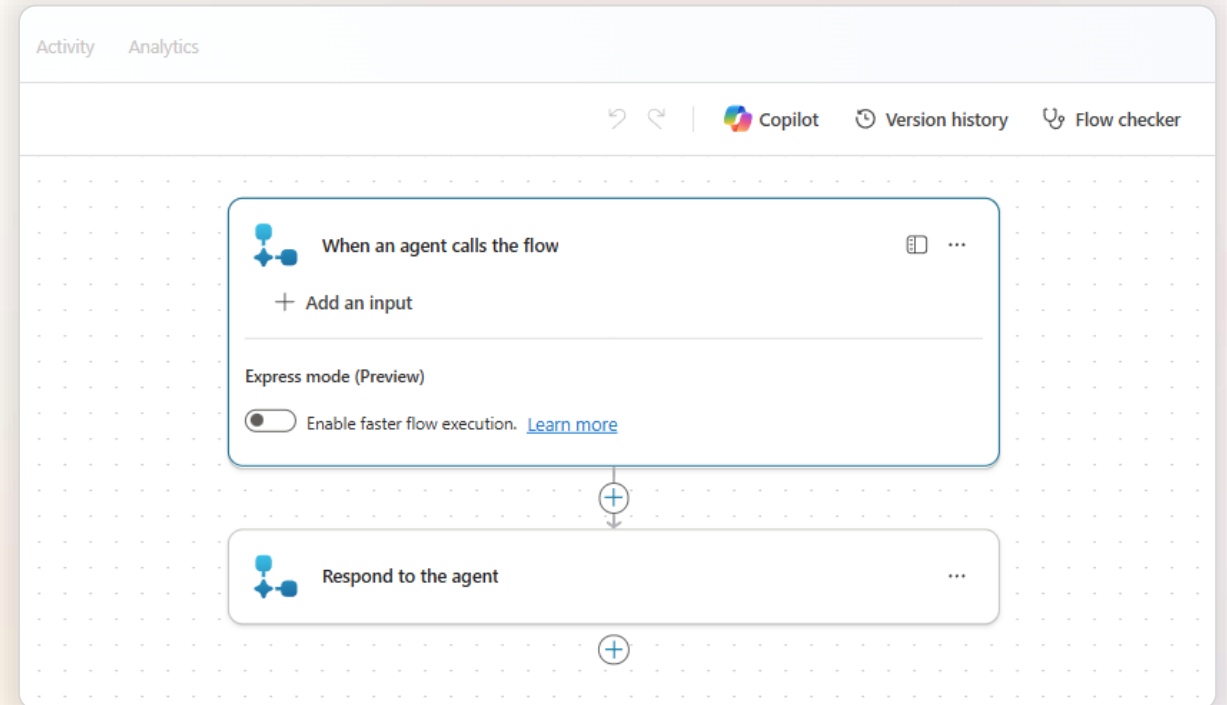


Create Copilot Studio workflows as tools

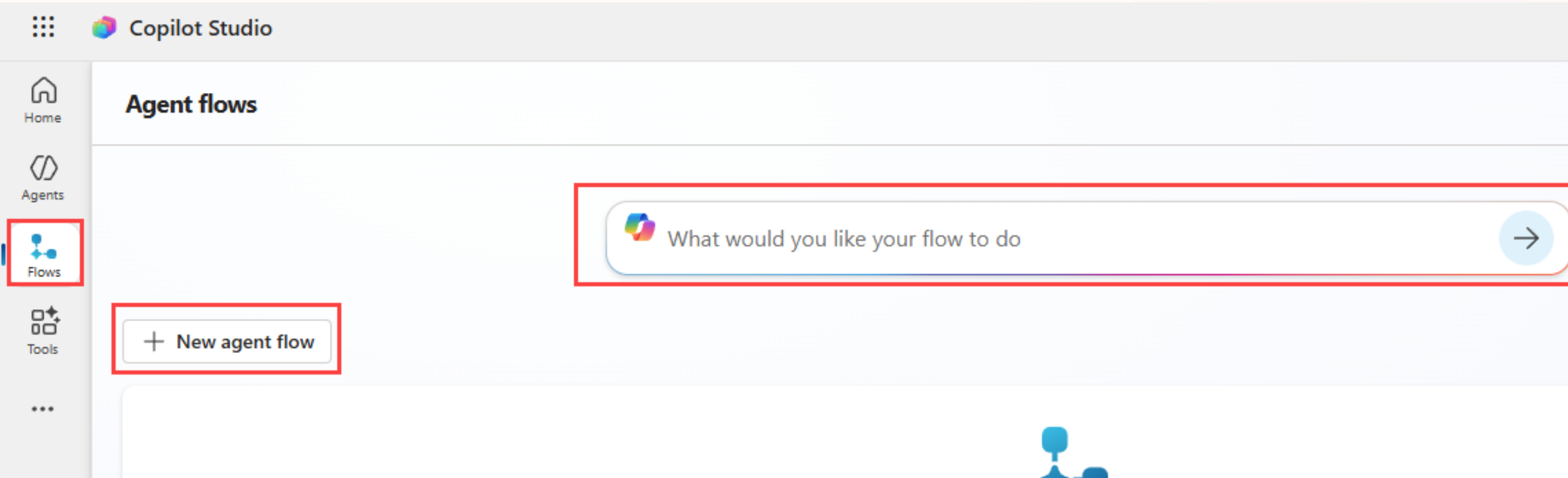
Copilot Studio agent flows as tools

Requirements for using as tools

- Use the **When an agent calls the flow** trigger
- Has the response action of **Respond to the agent**
- Configured to respond in real time, not asynchronously
- Must respond to the agent within the 100 second limit
- Must be in a Dataverse solution
- Must be published
- Must have a Copilot Studio plan associated with it
- Must only use these data types for parameters
 - String
 - Number
 - Boolean



Create an agent flow in Copilot Studio



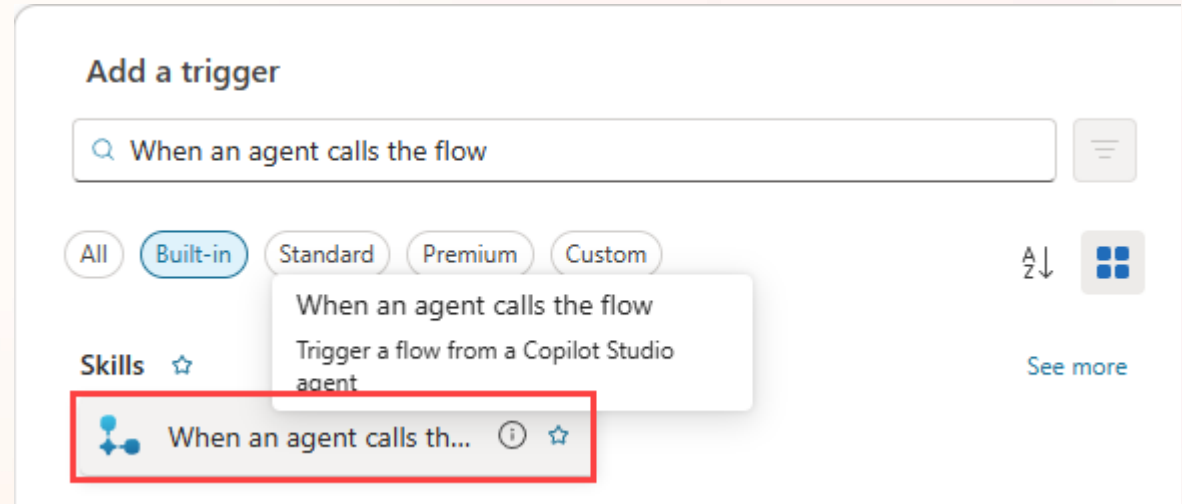
Designer: Build a workflow by selecting and adding components on a canvas in the visual designer

Natural language: Create a workflow by describing what you need using everyday language in one or more conversations

Create an agent flow in Copilot Studio

Trigger step

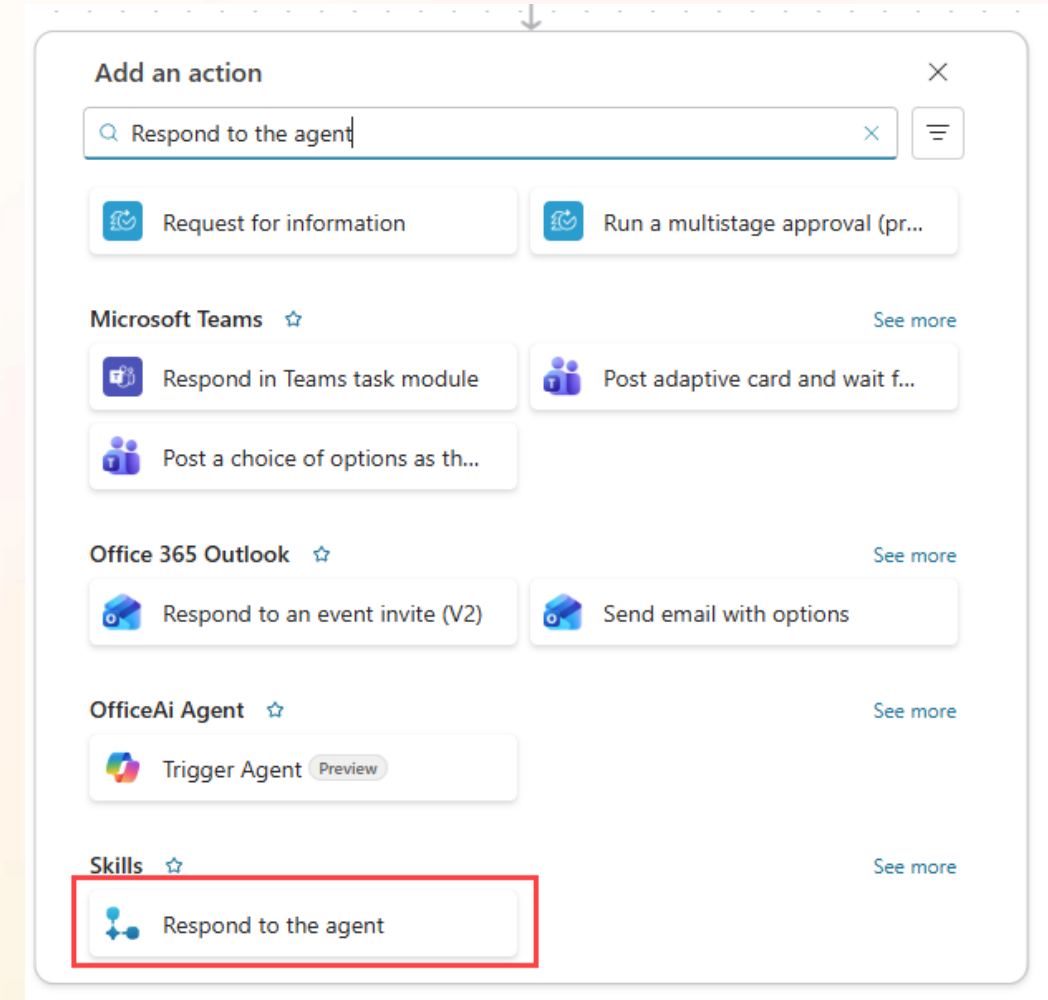
- **When an agent calls the flow** trigger
- Add **input parameters** to pass information from the agent to the workflow



Create an agent flow in Copilot Studio

Response action

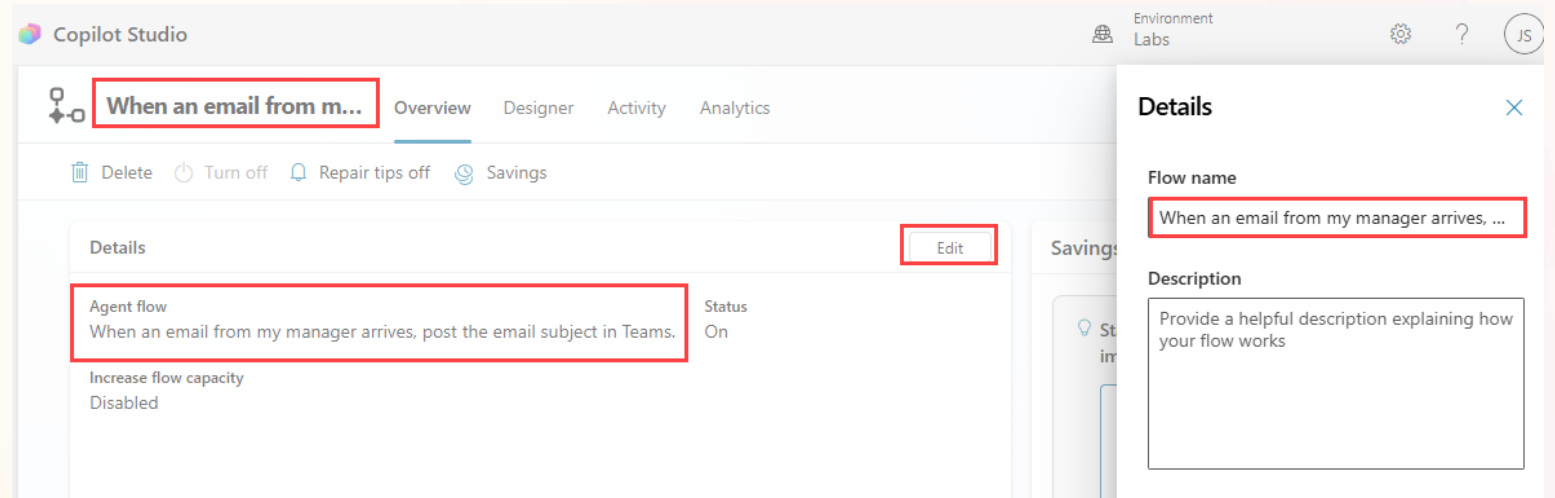
- **Respond to the agent** action
- Add **output parameters** to pass information from the workflow back to the agent
- Asynchronous response must be Off



Create an agent flow in Copilot Studio

Workflow name

- By default, when added as a tool the name of the workflow is the name of the tool
- Name the workflow to reflect its purpose and function
- The orchestrator will use this name when determining when to use the tool
- The name must be unique



Create a workflow as a tool in Copilot Studio

Tools registry

- Workflows are listed in the Tool registry in Copilot Studio
- Can create a workflow as a tool from the Tools registry

Name	Type
 Get Property	...
 Companies House	...
 <u>Companies House</u>	...
 Custom prompt	...
 Create Booking Request	...

Design Copilot Studio workflows

Design agent flows

Connectors

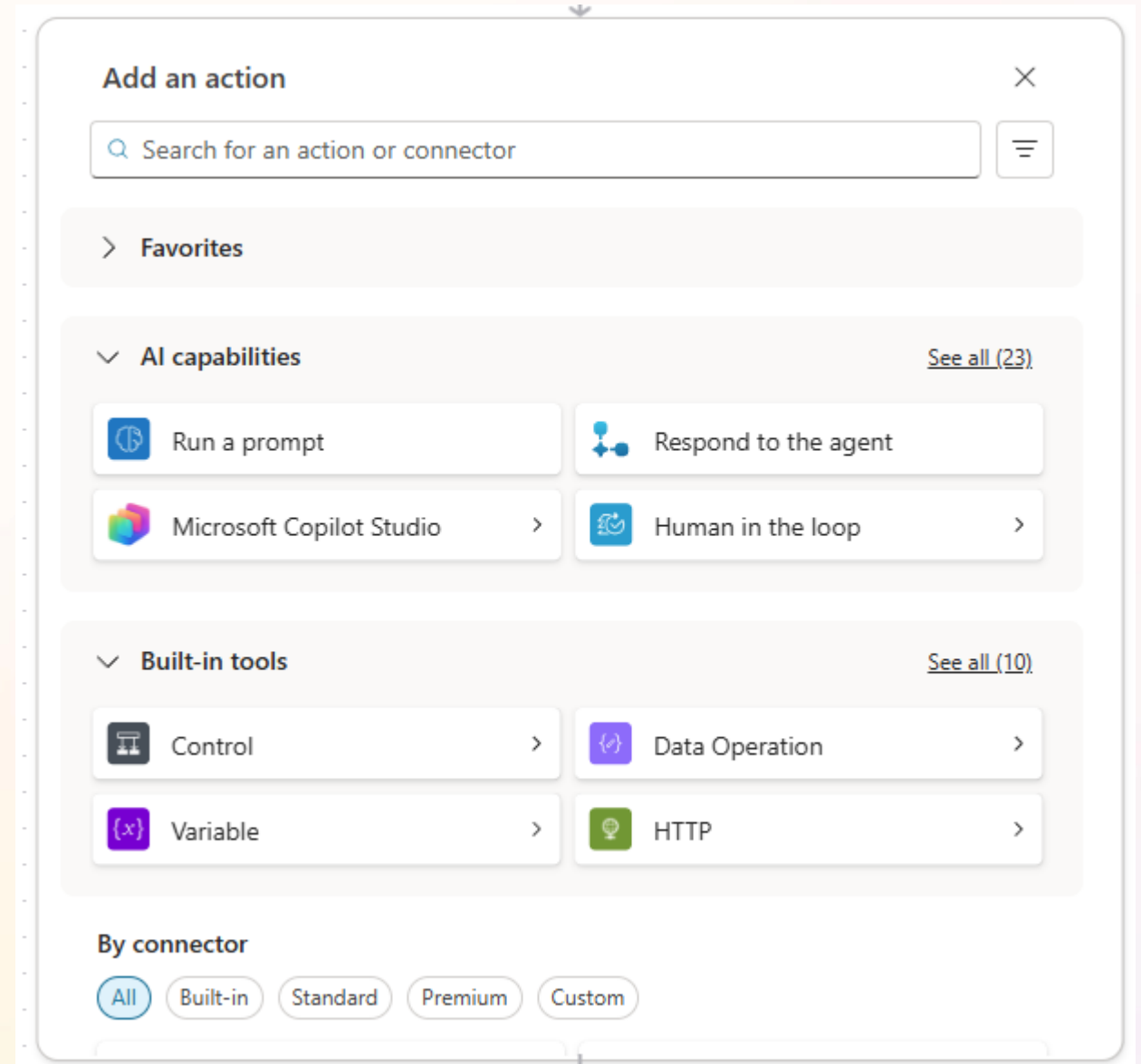
- Agent flows are like Power Automate cloud flows
 - Connectors
 - Dynamic content
 - Expressions
 - Conditions
 - Branches
 - Loops



Design agent flows

Add and configure actions

- Actions are steps that the agent follows or does
- Rename each action to make the workflow more readable and maintainable



Design agent flows

Dynamic content

- Reference data from the trigger and preceding actions
- Select an input field in the action
- Use Insert dynamic content icon or the / key
- Search for and select the data element to insert

The screenshot displays the Microsoft Power Automate interface for designing an agent flow. The main canvas shows three steps in a sequence:

- City**: A text input field with the placeholder "Enter the city".
- Country**: A text input field with the placeholder "Enter the country".
- Express mode (Preview)**: A toggle switch labeled "Enable faster flow execution. [Learn more](#)". Below it, a message says "Enjoying Express mode?" with thumbs up and down icons.

Below these steps, there are two more steps:

- Current time**: A step with a clock icon. A message below it says: "No additional information is needed for this step. You will be able to use the outputs in subsequent steps."
- Get forecast for today**: A step with a sun icon. A message below it says: "Invalid parameters". Below this, there is a "Location *" field.

On the right side, there is a "Parameters" panel with a search bar. It lists several parameters, including "Should the Peek Button Be Shown (msdyn_ShowPeekButton)", "Should only leaf node selection be allowed (msdyn_AllowSelectLeafOnly)", "SLA Web Client Deprecation Acknowledge (msdyn_SLAWebClientDeprecationAckn...", "LOB Guid List (msdyn_lineofbusinessfiltertemplatefeature)", and "LOB Guid List for user expansion (msdyn_lineofbusinessfiltertemplatefeatureusere...".

A red box highlights the "Current time" step in the main canvas and its corresponding entry in the "Parameters" panel. The "Current time" entry in the parameters panel shows the output "Current time" and the trigger "When an agent calls the flow". Below this, the inputs "City" (with placeholder "Enter the city") and "Country" (with placeholder "Enter the country") are listed.

Design agent flows

Expressions

- Used to
 - Transform data
 - Format data
 - Make decisions
- Select the ***fx*** icon within a step's input field
- Use Copilot to generate expressions



Create an expression with Copilot

Function Dynamic content

 Search

 String functions

concat(text_1, text_2?, ...)
Combines any number of strings together

substring(text, startIndex, length?)
Returns a subset of characters from a string.

slice(text, startIndex, endIndex?)
Returns a section of a string defined by the start index and the end index

replace(text, oldText, newText)
Replaces a string with a given string

guid()

Add

Design agent flows: Condition

The screenshot displays the Microsoft Power Automate interface for configuring a 'Condition' step. The left pane shows the 'Parameters' tab with a 'Condition expression' section. The right pane shows the 'Condition' step configuration with a table for 'Condition expression' and two output paths: 'True' and 'False'.

Condition expression configuration:

Value 1	Operator	Value 2
body/From	is equal to	Mail
Choose a value	is equal to	Choose a value
+ Add row		

Condition step configuration:

Condition expression *
Provide the values to compare and select the operator to use.

Value 1	Operator	Value 2
body/From	is equal to	Mail
Choose a value	is equal to	Choose a value
+ Add row		

True (Green button) → [+ Add](#) → **Post message in a chat or channel**

False (Red button) → **0 Actions**

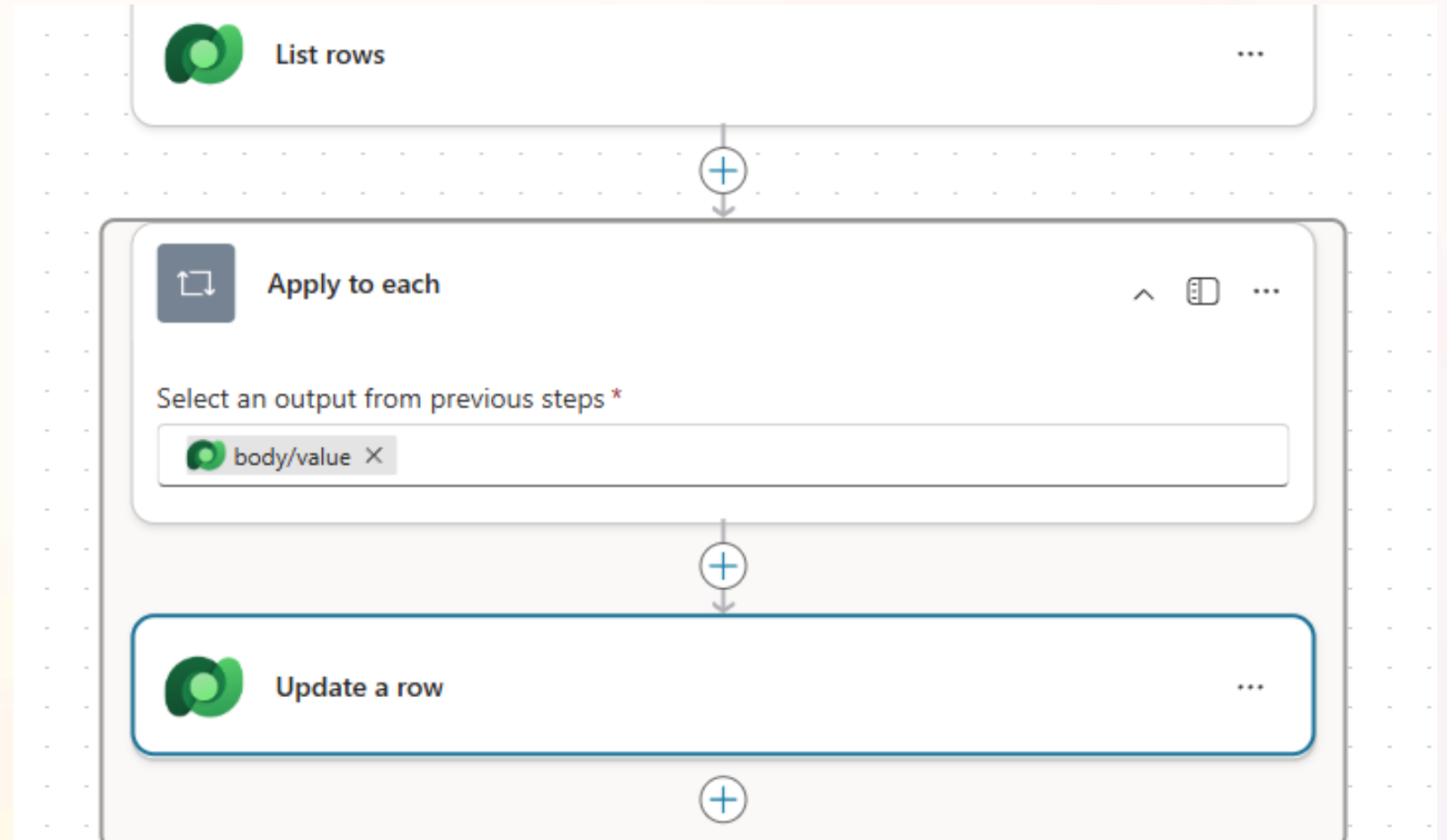
- Control connector
- If..then...else statement
- Adds a branch

- Use dynamic content and expressions in the left and right choose a value fields and select the operator

Design agent flows

Loops

- Apply for each
- Do until

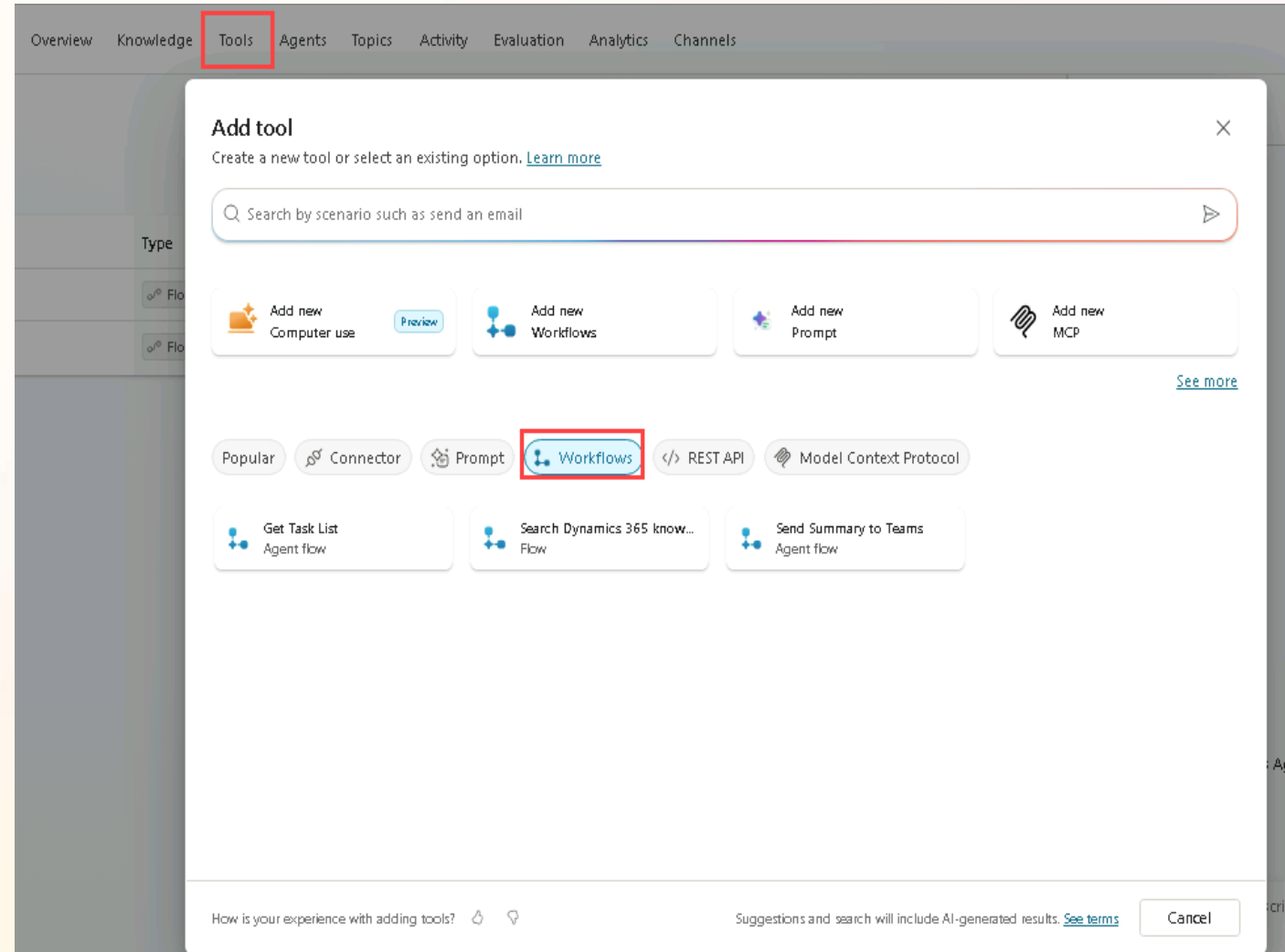


Add Copilot Studio workflows as tools to agents in Copilot Studio

Add Copilot Studio workflows as tools to agents

Add from Tools registry

- Tools tab in an agent
- Add a tool
- Filter by the Workflow icon
- Select the workflow



Add Copilot Studio workflows as tools to agents

Details

- Name
- Description
- Link to the workflow
- Additional details
 - When this tool may be used
 - Ask the end user before running

Details

Inputs

Completion

Details

What it is, how it operates, and how the orchestrator identifies it. [Learn more](#)

Name *

Get Today's Weather Forecast

28/64

Description * ⓘ

Get Today's Weather Forecast

28/1024

Agent flow

Get Today's Weather Forecast

Available to

Weather Forecast Assistant

⌵ Additional details

When this tool may be used ⓘ

Select ⌵

☒

Agent may use this tool at any time

☐

Only when referenced by topics or agents

Ask the end user before running

Recommended for sensitive or regulated domains or when making changes for the user

No ⌵

Add Copilot Studio workflows as tools to agents

Inputs

- Input parameters from the workflow are listed here
- Fill using:
 - Dynamically fill with AI (default)
 - Custom value

 **Inputs**

+ Add input

What the tool accepts in order to run. Inputs will be filled in the order shown.

Input name	Fill using ⓘ	Value	
City *	Dynamically fill with AI ▼	 Customize	⋮
Country *	Dynamically fill with AI ▼	 Customize	⋮

Add Copilot Studio workflows as tools to agents

Completion

- What will happen when the tool has finished
- After running
 - Don't respond (default)
 - Write the response with generative AI
 - Send specific response
 - Send an adaptive card
- Outputs from tool are listed

 **Completion**

Specify what your agent does when it finishes using this tool.

After running:

Don't respond (default) ▼

▼ **Advanced**

Outputs available to the agent and other tools

Defines which outputs are available to the agent and other tools

All ▼

Outputs

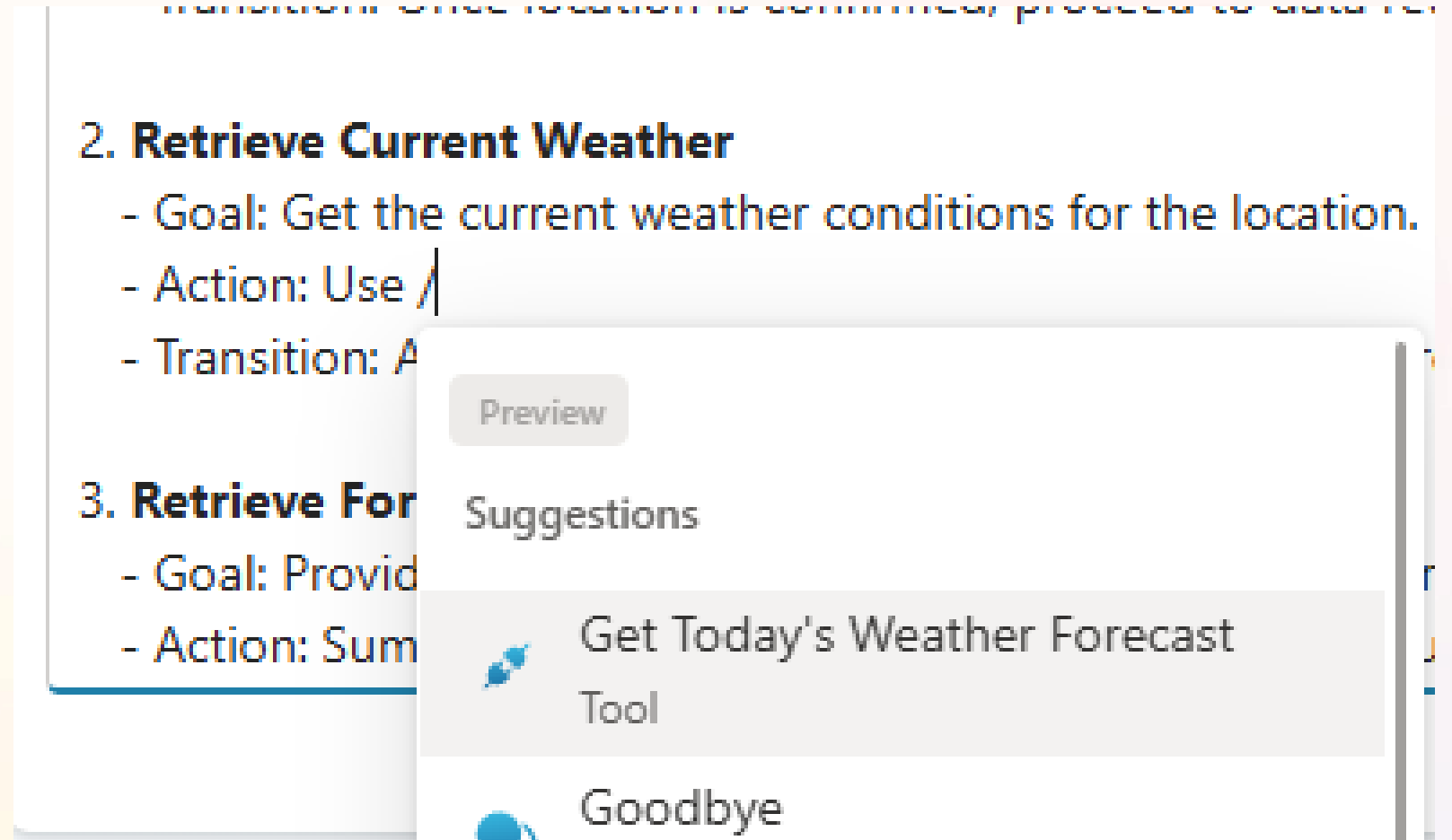
TodaySummary
todaysummary



Add Copilot Studio workflows as tools to agents

Agent instructions

- Generative AI selects one or more tools, topics, agents, and knowledge sources to create a response
- Agent instructions provide hints to the agent on what tools to select
- The tool and name and description are used by the agent
- Add a reference to the tool using / to direct the agent when to use the tool



Add Copilot Studio workflows as tools to agents

Create a new workflow tool for your agent

- Tools tab in an agent
- Add a tool
- Select the Workflows tile
- Both the workflow and tool will be named Untitled by default

Add tool

Create a new tool or select an existing option. [Learn more](#)

Search by scenario such as send an email



Add new
Computer use

Preview



Add new
Workflows



Add new
Prompt



Add new
MCP



Add new
REST API



Add new
Custom connector

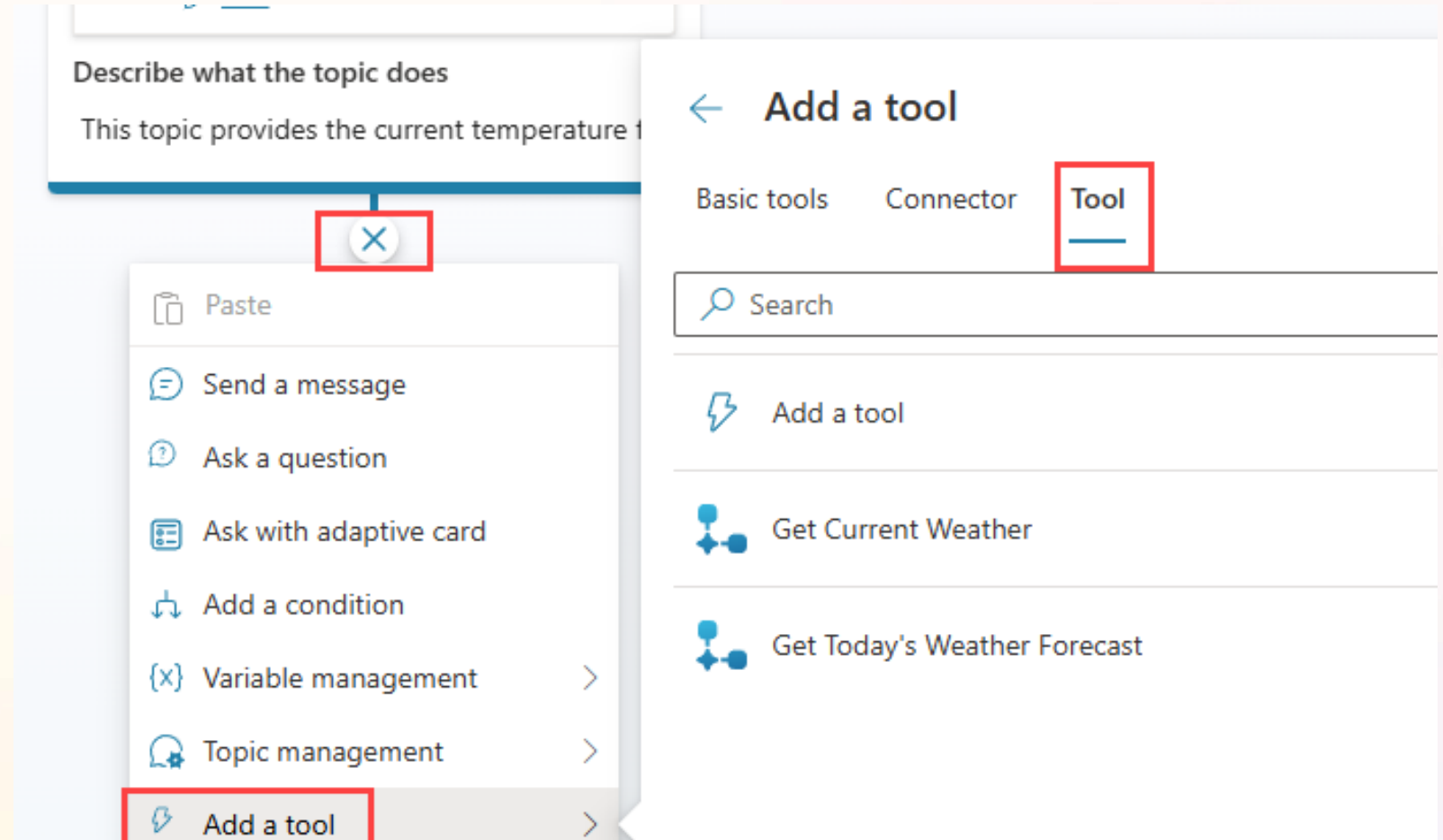
[See less](#)

Add Copilot Studio workflows as tools to topics in agents in Copilot Studio

Add Copilot Studio workflows as tools to topics

Using tools in topics

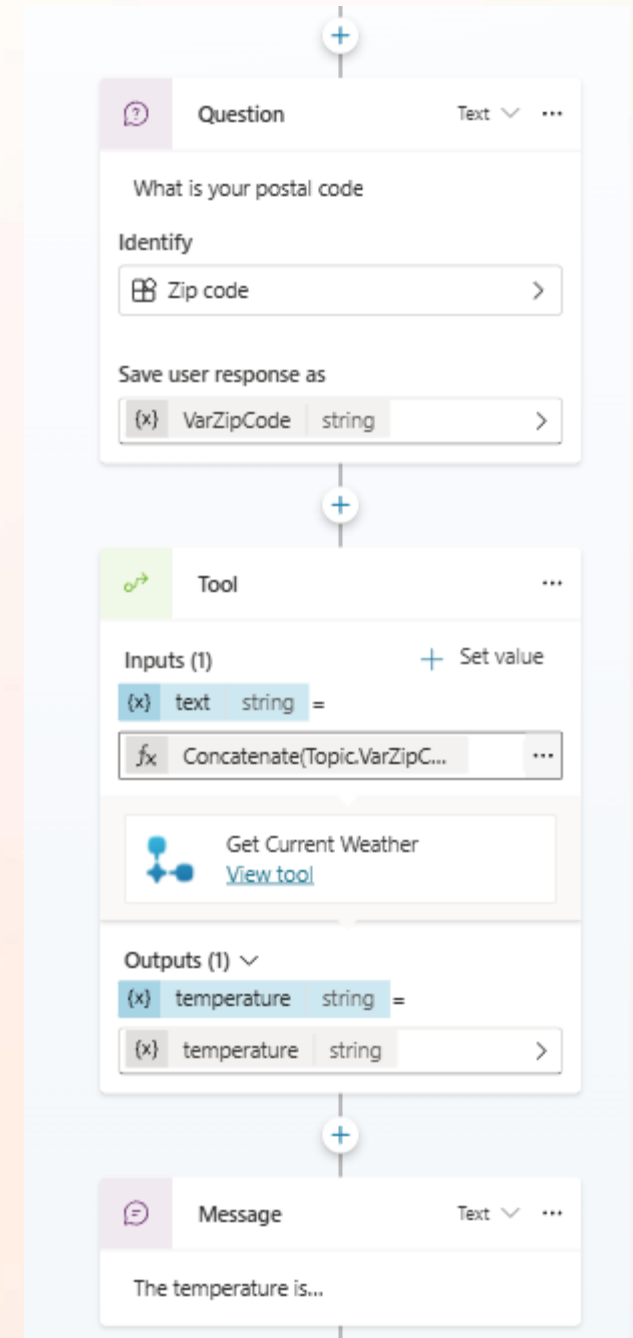
- Predictable execution:
- Strong control
- Reusability
- Reliability
- Maintainability



Add Copilot Studio workflows as tools to topics

Variables

- Use variables as input and output parameters to pass information between workflow tools and agents
- Add question nodes to capture the information required by the workflow



Practice Labs

Lab 4: Use workflows as tools

Practice Labs

Lab 5: Create an agent using the new Copilot Studio experience

Module Recap

In this module, we:

Learned how to create workflows in Copilot Studio for use as tools by agents

Learned how to add workflows as tools to agents in Copilot Studio

Learned how to use variables to pass information between workflows and agents



